BioLayers AI: Project History

Everything delivered from the first commit (1 Aug 2026) to 19 Oct 2026, compiled from the git history (102 commits, 522 files changed, about 164,600 lines added).

Summary

BioLayers AI is an AI-driven computational oncology platform. It turns research papers into evidence-linked mind maps, pairs them with a curated real-microscopy Cell Atlas, and validates a cell segmentation pipeline. Stack: Next.js (App Router), TypeScript, Tailwind, Supabase, deployed on Vercel, plus a Python research pipeline (Cellpose) for bioimage analysis.

Phase	Period	Focus
1. Launch	1-4 Aug	Initial site, founder section, Cell Atlas, local mode
2. Research workspace	7-19 Aug	v2 workspace, globe, knowledge graph, PubMed, evidence classification
3. Paper mind maps	19-23 Aug	Streaming mind-map pipeline, PDF parsing, model fallback chain
4. Accounts and platform	20-28 Aug	Supabase auth, Gemini BYOK, Library, usernames, analytics
5. Brand, press, partners	26 Aug-3 Sep	Press page, partners, mentors, executive PDF briefing
6. Project Cipher	2-3 Sep	Interactive cause-effect network and student decoder
7. Performance	2-4 Sep	Core Web Vitals, SSR, global background video
8. Cell Atlas expansion	12-29 Sep	Atlas foundation, modalities, 634-record catalog, CAF/CXCL12 series
9. Cell pipeline validation	1-2 Oct	Pipeline source, benchmarks, /validation page, hospital partnerships
10. Hardening	5-19 Oct	README, real-data content refresh, verified-science Cipher, team section

1. Launch (1-4 Aug)

- Initial BioLayers AI site; animation optimisation and founder layout fixes; founder section and photo
- Cell Atlas, local mode and UI improvements

2. Research workspace (7-19 Aug)

- BioLayers AI v2 research workspace
- Interactive planetary globe with country borders and flags; unified BioJourney
- Interactive cancer knowledge graph
- Evidence classification and improved PubMed research graph

- Homepage upgrade, UI and branding polish, performance work, homepage split into sections

3. Paper mind maps (19-23 Aug)

- Paper mind map with streaming pipeline, section-clustered layout and AI model fallback chain
- Replaced the canvas with a linear document reader in paper order
- PDF parsing fixes for Vercel (DOMMatrix crash, bundled pdf.worker)
- Fixed 413 on large uploads by extracting text client-side and sending JSON
- Clearer UX: mind map vs summarizer, Explain button, richer fallback graph entities
- Workspace redesign: circle nodes, compact toolbar, animated edges

4. Accounts and platform (20-28 Aug)

- Supabase authentication (PKCE) and per-user Gemini BYOK keys, encrypted with AES-256-GCM
- Saved papers reopen their mind map; marketing claims corrected to match shipped features
- Redesign around the dark-field fluorescence identity
- Vercel Analytics and Speed Insights; live platform stats on the home page
- Supabase linter fixes (RLS initplan, indexes, function grants)
- Library page with collective papers, user profiles and mind-map deduplication
- Unique username system, @username Library URLs, author attribution on mind maps
- Navbar and hero layout fixes; faster public pages (SSR hero, Supabase skipped on public routes)

5. Brand, press, partners and mentors (26 Aug-3 Sep)

- Press page with AI Business feature and real logo; HundrED global education innovation spotlight
- BioLayers logo and favicon set
- Partners: NXT Horizon, STEMise, CompBio Society at UCLA, BioMedizone; later Nationwide Children's Hospital and Ohio State University
- Mentors: Dr. John-William Sidhom, Dr. Eric E. Gardner, Dr. John Paul Shen, Dr. Kathleen Schieffer, Dr. Bailey Glen
- Executive PDF briefing; Ibn Sina school affiliation; contact emails updated

6. Project Cipher (2-3 Sep, refined 19 Oct)

- Interactive cause-effect network and student decoder
- Canvas battery optimisation (physics cooling, eco mode); fixed a 60fps re-render loop; stabilised movement; compact one-window layout
- Oct: verified science, evidence on every arrow, clearer graph, mobile layout

7. Performance (2-4 Sep)

- Core Web Vitals fixes, prerendered static routes, optimised auth proxy
- FCP and LCP improvements, framer-motion removed from the critical layout, image compression
- Global seamless background video with controls and reliable autoplay

8. Cell Atlas expansion (12-29 Sep)

- Phase 1 Foundation of the Cell Atlas; isolated wheel zoom
- More imaging modalities
- 634-record research microscopy catalog with provenance, and CAF/CXCL12 migration tracking series
- Frontend visual system redesign

9. Cell pipeline validation (1-2 Oct)

- Cell pipeline source (research/cell-pipeline), benchmark reports, result figures, demo video, per-cell JSON and PNG overlays (AI/ML workstream, merged via PR #2)
- /validation page rendering the validation report; navbar and footer links; click-to-zoom figures
- Reported results: BBBC039 IoU 0.942 / F1 0.972; U373 SEG 0.940
- Nationwide Children's Hospital and Ohio State University partnerships; all partner logos displayed

10. Hardening and team (5 Oct onward)

- README with platform documentation
- Content refreshed with real data; dead code and an unrelated project removed
- Plan-vs-state audit and team task briefs (docs/TEAM_TASKS.md, docs/tasks)
- Team section on the About page (7 members)

Team

Name	Role
Azizbek Gayratov	Founder & Marketing
Bilolbek Yuldoshkhujayev	Software Engineer
Asliddin Abdurakhimov	Data Analyst & AI/ML Engineer
Orzugul Mukhtorova	Data Analyst
Muhammad Najmiddinov	AI/ML Engineer
Maksat Babageldiyev	AI/ML Engineer
Mirkomil Nozimkhojiyev	SMM

Open items (from the 19 Oct audit)

- Verify the 634 catalog records against real image files; only 6 images ship in public/atlas
- CTC and CC-BY-NC dataset licensing review
- Phenotype labels are rule-based and not expert-validated
- Segmentation runs offline in Python, not inside the web product
- Social reach is far below targets; no researcher demo feedback yet